

Relational Dualities and Bisimulation

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Abstract

The Kripke semantics of various logics arise via categorical dualities between a category of relational frames and their maps, and a category of algebras and logical homomorphisms. When the relational frames are considered as computational systems (e.g. the states of a machine), the corresponding algebra is one of logical predicates on these systems (e.g. predicates on these states, i.e. program logics). Our aim is to extend this phenomenon to relations, putting well-behaved relations between systems (e.g. bisimulations) in correspondence with relations between predicates. This is achieved by constructing particular relational extensions of Tarski duality (for infinitary classical propositional logic) and Thomason duality (for infinitary classical modal logic). We sketch how these dualities give rise to a proof system that relates formulae between different systems.

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1 Introduction

It is a well-known secret that relational semantics (e.g. Kripke semantics) arise from *dualities*, i.e. categorical equivalences of the form $F : \mathcal{C}^{\text{op}} \simeq \mathcal{D}$. The idea is that $\mathcal{C}$ is a ‘spatial’ category whose objects are *frames*, and $\mathcal{D}$ is an ‘algebraic’ category whose objects are *algebras of predicates*, which allow reasoning about the elements of the frames.

For example, truth table semantics for infinitary classical propositional logic arise from the *Tarski duality* $\mathbf{Set}^{\text{op}} \simeq \mathbf{CABA}$ between the category of sets and the category of *complete atomic Boolean algebras* (CABAs) [21, §1].¹ The Kripke semantics of infinitary classical modal logic [22, 24] arise from the *Thomason duality* $\mathbf{Frm}_{\text{open}}^{\text{op}} \simeq \mathbf{CABAO}^-$ between the category of Kripke frames and the category of CABAs with operators (i.e. modalities) [38] [21, §2.4]. The Kripke semantics of infinitary intuitionistic logic [23] arises from the duality $\mathbf{Pos}^{\text{op}} \simeq \mathbf{PrAlgLatt}$ between posets and prime algebraic lattices [31, 20]. Famous dualities of this form include Stone duality (for Boolean algebras) [18], Jónsson-Tarski duality [19, 11, 33] (for Boolean algebras with operators), and Esakia duality [8] (for Heyting algebras). See the chapter by Kishida [21] for further references.

The overwhelming majority of work in this area has not paid much attention to the logical rôle of *morphisms* of these categories.² In many cases the ‘frame’ morphisms $f : W \rightarrow W'$

¹ The semantics of classical propositional logic arises from the duality $\mathbf{Set}_{\text{fin}}^{\text{op}} \simeq \mathbf{Bool}_{\text{fin}}$ between finite sets and finite Boolean algebras. In fact, we can arrive at the Tarski duality $\mathbf{Set}^{\text{op}} \simeq \mathbf{CABA}$ by taking the Ind-completion of this duality [26].

² One exception is the work of Abramsky [1] on domain theory in logical form.



of $\mathcal{C}$ can be thought of as ‘truth-preserving’: they map worlds to worlds in a way that preserves and reflects truth, i.e. $w \models \varphi$ implies $f(w) \models \varphi$. The corresponding morphism $f^* : F(W') \rightarrow F(W)$ of $\mathcal{D}$ then preserves the logical structure, e.g. $f^*(\varphi \wedge \psi) = f^*(\varphi) \wedge f^*(\psi)$. Thus f^* can be considered as mapping predicates over the W to predicates over W' , in a manner that preserves logical connectives, and hence reasoning.

Obtaining this property in the case of intuitionistic and modal logics requires strong properties on frame morphisms. For example, in the Kripke semantics of modal logic the induced map $f^* : \mathcal{P}(W') \rightarrow \mathcal{P}(W)$ preserves the modality $\Box$ iff $f : W \rightarrow W'$ is an *open map*, i.e. a transition-preserving function whose graph is a (*functional*) *bisimulation* [21, §2.4].³

This observation raises an interesting question: what happens if we relax the morphisms of $\mathcal{C}$, requiring them to merely be *relations* instead of functions? Answering this question would allow us to have categories $\mathcal{C}$ consisting of frames and well-behaved relations, e.g. bisimulations. A duality of this form would then put bisimulations in correspondence with morphisms between algebras, enabling some kind of reasoning ‘across’ a bisimulation.

Such results, which may be called *relational dualities*, are already found in the literature. However, most of them put relations between frames in correspondence with *hemimorphisms*, i.e. morphisms that preserve most—but not all!—logical connectives. For example, a relation $R \subseteq W \times W'$ corresponds to a function $W \rightarrow \mathcal{P}(W')$, which uniquely corresponds to a function $R^* : \mathcal{P}(W') \rightarrow \mathcal{P}(W)$ which preserves all joins (but not necessarily all meets) [19] [21, §2.3]. This is unsatisfactory from a computational perspective: traditional dualities allow us to compute the ‘action’ of a morphism of frames $f : W \rightarrow W'$ on the *syntax* of a formula: given a formula φ (over W') we can define a formula $f^*(\varphi)$ (over W) by recursion, preserving connectives and replacing each proposition p with a proposition $f^{-1}(p)$ which is true at w exactly when p is true at $f(w)$. As hemimorphisms do *not* preserve all logical connectives, it is impossible to adapt this pattern of definition to a relation $R \subseteq W \times W'$.

To tackle this difficulty this paper develops a different type of duality, viz. one that puts relations $R \subseteq W \times W'$ between frames in correspondence with relations $\mathcal{F}(R) \subseteq F(W) \times F(W')$ between predicates. This allows us to view $\mathcal{F}(R)$ as a *relational judgment* between formulae pertaining to different systems, with R playing the rôle of a ‘background theory’ which specifies how propositions over the two systems are related. The ultimate goal is to develop a formal system for synthetically reasoning about the *relation between formulae over different frames*. It is not difficult to imagine that this might have applications in program logics.

If the relation $R \subseteq W \times W'$ is also a *bisimulation* between frames, it can be thought of as a notion of equivalence between the states of two systems, with $w R w'$ meaning that w and w' have the same behaviours. Viewed as an judgment, the induced relation $\mathcal{F}(R) \subseteq F(W) \times F(W')$ then satisfies additional ‘inference rules.’

As we are interested in bisimulations, we take a hint from the relevant literature on coalgebra [37, 28, 17] and base all the results in this paper on the *lower lifting*, which maps a relation $R \subseteq W \times W'$ to the *lower relation* $\mathcal{L}(R) \subseteq \mathcal{P}(W) \times \mathcal{P}(W')$, defined by

$$S \mathcal{L}(R) T \stackrel{\text{def}}{=} \forall s \in S. \exists t \in T. s R t.$$

Famously, this construction is *not* a functor $\mathbf{Rel} \rightarrow \mathbf{Rel}$, as it does not preserve identities. However, we can define an appropriate category **CABARel** whose objects are CABAs, so that it becomes a functor $\mathcal{L} : \mathbf{Rel} \rightarrow \mathbf{CABARel}$. Surprisingly, this functor is an

³ Open maps are also called *p-morphisms* [4, §2.3] or *bounded morphisms* [3, §2.1].

equivalence. Moreover, it ‘extends’ Tarski duality, in the sense that the diagram

$$\begin{array}{ccc}
 \mathbf{Set}^{\text{op}} & \xrightarrow{\quad} & \mathbf{Rel}^{\text{op}} \\
 \simeq \downarrow & & \simeq \downarrow \mathcal{L} \circ (-)^{\dagger} \\
 \mathbf{CABA} & \xrightarrow{\quad} & \mathbf{CABARel}
 \end{array}$$

commutes, where $(-)^{\dagger} : \mathbf{Rel}^{\text{op}} \simeq \mathbf{Rel}$ is the obvious formal duality that reverses a relation. This follows a known pattern whereupon relational dualities are not really ‘dualities’ per se, but become so only after composition with a formal duality [21, §2.3] [25, Remark 4.1]. In this light the $(-)^{\text{op}}$ is merely an artifact of functionality.

After recalling some preliminary definitions to fix notation (Section 2), we briefly recap the elements of the Tarski duality $\mathbf{Set}^{\text{op}} \simeq \mathbf{CABA}$ (Section 3). Then, we show the lower lifting given above can be used to extend it to a *relational Tarski duality* (Section 4). We achieve this by precisely characterising the relations between CABAs that are obtained as lower liftings, and hence establish an equivalence. Furthermore, we show that this is an extension of the usual Tarski duality, and sketch the formal system that this duality induces.

Following this, we show how the same lifting gives rise to a relational version of the Thomason duality $\mathbf{Frm}_{\text{open}}^{\text{op}} \simeq \mathbf{CABAO}$ (Section 5). This is somewhat more challenging: while the Tarski duality acts on arbitrary functions, the Thomason duality associates open maps (i.e. functional bisimulations) with modal complete Boolean homomorphisms. We show that this has an analogue on the relational level, putting (bi)simulations in unique correspondence with what we call *(bi)simulatory relations* between CABAs. We show that this *relational Thomason duality* extends the duality originally given by Thomason, and again sketch the formal system that this duality induces.

2 Preliminaries

A *relation* $R : A \rightarrowtail B$ from a set A to a set B is a subset $R \subseteq A \times B$. We write $a R b$ to mean $(a, b) \in R$. If $R : A \rightarrowtail B$ and $S : B \rightarrowtail C$ their composition $R ; S : A \rightarrowtail C$ is defined in the usual manner, i.e. $a (R ; S) c$ just if there exists some $b \in B$ with $a R b S c$. Note that we write the composition of relations using diagrammatic order. Given a relation $R : A \rightarrowtail B$ we define its *opposite* $R^{\dagger} : B \rightarrowtail A$ by $b R^{\dagger} a$ iff $a R b$. Sets and relations form a category $\mathbf{Rel}$ under composition, with the identity $\text{Id}_A : A \rightarrowtail A$ given by $a_1 \text{Id}_A a_2$ iff $a_1 = a_2$. Moreover, the opposite construction extends to a functor $(-)^{\dagger} : \mathbf{Rel}^{\text{op}} \rightarrow \mathbf{Rel}$ which induces a formal duality $\mathbf{Rel}^{\text{op}} \simeq \mathbf{Rel}$ given by taking the opposite of a relation. This shows that $\mathbf{Rel}$ is a *dagger category* [13].

A partial order $(P, \sqsubseteq_P)$ is a set P equipped with a relation $\sqsubseteq_P : P \rightarrowtail P$ which is reflexive, transitive, and anti-symmetric. We will commonly refer to partial orders by their carriers $(P, Q, \dots)$ and often elide the subscript of $\sqsubseteq$. A function $f : P \rightarrow Q$ is *monotonic* just if $x \sqsubseteq_P y$ implies $f(x) \sqsubseteq_Q f(y)$. A *lattice* is a partial order that has all finite joins and meets. We will use $\mathcal{L}, \mathcal{L}', \dots$ to denote lattices. A *complete lattice* $\mathcal{L}$ is a partial order that has all joins, which also implies that it has all meets. A pair of monotonic functions $f : P \rightarrow Q$ and $g : Q \rightarrow P$ is an *adjoint pair* (or *adjunction*, and in this setting also called a *Galois connection*) just if $f(x) \sqsubseteq_Q y$ iff $x \sqsubseteq_P g(y)$. This is often denoted as $f \dashv g$. It is a consequence that f preserves all joins and g preserves all meets. The *adjoint functor theorem* [6, §7.34] [18, §I.4.2] says that if $f : P \rightarrow Q$ is monotonic, P is a complete lattice, and f preserves all joins, then f has a right adjoint $g : Q \rightarrow P$. Dually, if $g : Q \rightarrow P$ is monotonic, Q is a complete lattice, and g preserves all meets, then g has a left adjoint $f : P \rightarrow Q$.

We refer the reader to [6] for further background in orders and lattices.

3 A Primer on Tarski Duality

We recap the Tarski duality $\mathbf{Set}^{\text{op}} \simeq \mathbf{CABA}$, and show how it induces an (infinitary) classical propositional logic. Recall that a *Boolean algebra* is a distributive lattice $\mathcal{B}$ in which every element $x \in \mathcal{B}$ has a *complement* $\neg x \in \mathcal{B}$ for which $x \vee \neg x = \top$ and $x \wedge \neg x = \perp$ [18].

For any set X , its powerset $\mathcal{P}(X)$ is a Boolean algebra [6, §5.2]. In fact, it is a *complete* Boolean algebra, i.e. a complete lattice with joins and meets of $(S_i)_{i \in I}$ given by $\bigcup_{i \in I} S_i$ and $\bigcap_{i \in I} S_i$ respectively. Moreover, $\mathcal{P}(X)$ is *atomic*, in the following sense.

► **Definition 1 (Atom).** Let $(P, \sqsubseteq)$ be a partial order with a bottom element $\perp$. An element $a \in P$ is an *atom* of P if $a \neq \perp$ and $x \sqsubseteq a$ implies that either $x = \perp$ or $x = a$. We write $\text{At}(P)$ for the set of atoms of P .

► **Definition 2 (Atomic complete lattice).** A complete lattice $\mathcal{L}$ is *atomic* just if every element is equal to the join of atoms below it, i.e. for every $x \in \mathcal{L}$ we have $x = \bigsqcup \{a \in \text{At}(\mathcal{L}) \mid a \sqsubseteq x\}$.

A *complete atomic Boolean algebra (CABA)* is then a complete Boolean algebra $\mathcal{B}$ which is moreover atomic. For any set X , the powerset $\mathcal{P}(X)$ is a CABA: its atoms are exactly the singleton sets $\{x\}$ for $x \in X$; every $S \subseteq X$ can be reconstructed as $S = \bigcup_{x \in S} \{x\}$.

A *morphism of CABAs* $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ is a monotonic function that preserves all meets and joins. This definition has deceptively many implications. First, as $\mathcal{B}$ and $\mathcal{B}'$ are Boolean algebras, every such $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ is a *complete Boolean homomorphism*, i.e. preserves *all* Boolean connectives, including negation. Second, by the adjoint functor theorem the map $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ has both a left and a right adjoint:

$$\begin{array}{ccc} & f_! & \\ \mathcal{B}' & \begin{array}{c} \xleftarrow{\perp} \\ \xleftarrow{f^*} \\ \xleftarrow{\perp} \end{array} & \mathcal{B} \\ & f_* & \end{array}$$

CABAs and their morphisms form a category **CABA**.

Given any function $f : X \rightarrow Y$ we can define a complete Boolean homomorphism $f^* : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ by $f^*(B) = \{x \in X \mid f(x) \in B\}$ [27, 30, 21]. By the adjoint functor theorem, f^* has both a left and a right adjoint, given by

$$\begin{aligned} f_!(A) &= \{y \in Y \mid \exists x \in X. f(x) = y \wedge x \in A\} = \{f(x) \mid x \in A\} \\ f_*(A) &= \{y \in Y \mid \forall x \in X. f(x) = y \Rightarrow x \in A\} = \{y \in Y \mid f^*({x}) \subseteq A\} \end{aligned}$$

Thus, we obtain a functor $\mathcal{P} : \mathbf{Set}^{\text{op}} \longrightarrow \mathbf{CABA}$.

In order to invert a functor we need to use some standard properties of these left and right adjoints [9, Lemma 1.23 and Exercise 1.3.10.e].

► **Definition 3 (Prime).** An element j of a complete lattice $\mathcal{L}$ is *prime* just if for any subset $S \subseteq \mathcal{L}$, if $j \sqsubseteq \bigvee S$ then there exists some $s \in S$ for which $j \sqsubseteq s$.

► **Lemma 4.** Let $f^* : \mathcal{L} \rightarrow \mathcal{L}'$ be a complete lattice homomorphism between complete lattices. Then its left adjoint $f_! : \mathcal{L}' \rightarrow \mathcal{L}$ preserves primes.

► **Lemma 5.** In a CABA $\mathcal{B}$ atoms and primes coincide.

► **Corollary 6.** Let $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ be a morphism of CABAs. Then its left adjoint $f_! : \mathcal{B}' \rightarrow \mathcal{B}$ maps atoms to atoms.

We can thus define a functor $\text{At} : \mathbf{CABA} \rightarrow \mathbf{Set}^{\text{op}}$, which maps a CABA $\mathcal{B}$ to its set of atoms $\text{At}(\mathcal{B})$, and use Corollary 6 to map a morphism $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ to the restriction $f!|_{\text{At}(\mathcal{B})} : \text{At}(\mathcal{B}) \rightarrow \text{At}(\mathcal{B}')$ of its left adjoint to atoms.

This is a pseudo-inverse to $\mathcal{P}$, and we obtain a categorical equivalence.

► **Theorem 7** (Tarski duality). $\mathbf{Set}^{\text{op}} \simeq \mathbf{CABA}$

The gist is that every CABA $\mathcal{B}$ is isomorphic to the powerset $\mathcal{P}(\text{At}(\mathcal{B}))$ of its atoms: its elements are uniquely determined by the atoms below them.

This duality induces (infinitary) classical propositional logic. Let W be a set of *worlds*. This set corresponds to the powerset $\mathcal{P}(W)$, which can be seen as a set of *predicates*. A world $w \in W$ satisfies the predicate $\varphi \in \mathcal{P}(W)$ just if $w \in \varphi$. As $\mathcal{P}(W)$ is a Boolean algebra, predicates are closed under all Boolean operations.⁴ Individual worlds $w \in W$ correspond to singleton predicates $\{w\} \in \mathcal{P}(W)$ which uniquely characterise them. If we interpret W as the *set of states* of a computer, a logic like this has been in continuous use since the pioneering work of Dijkstra [7].

4 A Relational Tarski Duality

If instead of a function $f : W \rightarrow W'$ we were to have an arbitrary *relation* $R : W \rightarrowtail W'$, what would the corresponding morphism between $\mathcal{P}(W)$ and $\mathcal{P}(W')$ be?

It is possible to obtain such a duality by replacing $\mathbf{CABA}$ with a larger category $\mathbf{CABA}_{\vee}$, whose morphisms only preserve joins, and then show that $\mathbf{Rel} \simeq \mathbf{CABA}_{\vee}$ [19, 35] [21, §2.3]. Indeed, if $h : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$ preserves all joins, then, writing any set as a union of singletons, we see that h is completely determined by a function $X \rightarrow \mathcal{P}(Y)$, i.e. a relation $X \rightarrowtail Y$. Hofmann and Nora [15, §4.5] show that this duality is a special case of a general monadic one. However, these more general morphisms do *not* preserve the logical structure of predicates. For example it may be that $h(\varphi \wedge \psi) \neq h(\varphi) \wedge h(\psi)$. Thus, it is no longer possible to compute h on the syntax of predicates, as we cannot straightforwardly define it recursively. We have therefore lost the benefit of presenting the morphism h as a function.

Instead, we would like a duality whose logical side consists of predicates and *relations*. To construct one we consider a simple way of *lifting* relations to powersets.

► **Definition 8** (Lower Relation). *Given a relation $R : X \rightarrowtail Y$, the associated lower relation $\mathcal{L}(R) : \mathcal{P}(X) \rightarrowtail \mathcal{P}(Y)$ is defined by*

$$S \mathcal{L}(R) T \stackrel{\text{def}}{=} \forall s \in S. \exists t \in T. s R t.$$

This lifting has a long history. It is one half of the *Egli-Milner lifting*, which first appeared in the powerdomain literature [32, 12]. It then resurfaced in the coalgebra literature [37], where post-fixed-points amount to *simulations* between transition systems [17, §3]. Intuitively, $X \mathcal{L}(R) Y$ means every state of X can be ‘ R -simulated’ by a state of Y .

$\mathcal{L}$ maps a relation between sets to a relation between CABAs. However, it is *not* a functor $\mathbf{Rel} \rightarrow \mathbf{Rel}$: it maps the identity relation $\text{id} : X \rightarrowtail X$ to the containment relation $\subseteq : \mathcal{P}(X) \rightarrowtail \mathcal{P}(X)$. We will therefore define a category $\mathbf{CABARel}$ so that $\mathcal{L}$ becomes a functor $\mathbf{Rel} \rightarrow \mathbf{CABARel}$. The relations between CABAs will be characterised as follows.

⁴ The fact $\mathcal{P}(W)$ is a *complete* Boolean algebra means that predicates are also closed under *infinite* conjunction $\bigwedge_{i \in I} \varphi_i$ and disjunction $\bigvee_{i \in I} \varphi_i$.

► **Definition 9** (Directionally atomic relation). Let P, Q be partial orders, $\mathcal{L}$ be a complete lattice, and $\mathcal{B}, \mathcal{B}'$ be CABAs.

1. A relation $R : P \rightarrowtail Q$ is a bimodule just when $p' \sqsubseteq p R q \sqsubseteq q'$ implies $p' R q'$.
2. A relation $R : \mathcal{L} \rightarrowtail Q$ is left-disjunctive just if $a_i R b$ for all $i \in I$ implies $(\bigsqcup_{i \in I} a_i) R b$.
3. A relation $R : \mathcal{B} \rightarrowtail \mathcal{B}'$ is atomic-founded when for any atom $a \in \mathcal{B}$, if $a R b$ then there exists an atom $b' \sqsubseteq b$ with $a R b'$.
4. A relation $R : \mathcal{B} \rightarrowtail \mathcal{B}'$ is directionally atomic if it is left-disjunctive, atomic-founded, and a bimodule.

It is simple to show that directionally atomic relations compose, and that $\sqsubseteq$ is the identity.

► **Proposition 10.** If $R : \mathcal{B} \rightarrowtail \mathcal{B}'$ and $S : \mathcal{B}' \rightarrowtail \mathcal{B}''$ are directionally atomic, then so is their composition $R ; S : \mathcal{B} \rightarrowtail \mathcal{B}''$.

► **Proposition 11.** $\sqsubseteq : \mathcal{B} \rightarrowtail \mathcal{B}$ is directionally atomic, and is the identity for composition of directionally atomic relations.

We can therefore define a category **CABARel** with CABAs as objects, and directionally atomic relations as morphisms. We can then construct a functor:

► **Lemma 12.** $\mathfrak{L}(R) : \mathcal{P}(X) \rightarrowtail \mathcal{P}(Y)$ is directionally atomic for any relation $R : X \rightarrowtail Y$.

► **Lemma 13.** $\mathfrak{L}$ is a functor $\mathbf{Rel} \rightarrow \mathbf{CABARel}$.

Proof. By Lemma 12 every $\mathfrak{L}(R)$ is a morphism in **CABARel**.

First, we show that $\mathfrak{L}$ preserves the identity. If $\text{id} : X \rightarrowtail X$ then $\mathfrak{L}(\text{id})$ is just the relation $\sqsubseteq$, which is the identity $\mathcal{P}(X) \rightarrowtail \mathcal{P}(X)$ in **CABARel**.

Second, we show that $\mathfrak{L}$ preserves composition. Let $R : X \rightarrowtail Y$ and $S : Y \rightarrowtail Z$ be relations. Suppose $A \mathfrak{L}(R ; S) C$. For any $a \in A$ define

$$B_a \stackrel{\text{def}}{=} \{b_a \in Y \mid a R b_a \wedge \exists c \in C. b_a S c\}, \quad B \stackrel{\text{def}}{=} \bigcup_{a \in A} B_a$$

Then $A \mathfrak{L}(R) B$ and $B \mathfrak{L}(S) C$, so $A \mathfrak{L}(R) ; \mathfrak{L}(S) C$.

For the converse, suppose instead that $A \mathfrak{L}(R) ; \mathfrak{L}(S) C$, so there exists some B with $A \mathfrak{L}(R) B$ and $B \mathfrak{L}(S) C$. Then for every $a \in A$ there must exist $b \in B$ and $c \in C$ with $a R b S c$. Hence $A \mathfrak{L}(R ; S) C$. ◀

► **Lemma 14.** $\mathfrak{L} : \mathbf{Rel} \rightarrow \mathbf{CABARel}$ is faithful.

Proof. Suppose $R, S : X \rightarrowtail Y$ are two relations and that $\mathfrak{L}(R) = \mathfrak{L}(S)$. Then $x R y$ iff $\{x\} \mathfrak{L}(R) \{y\}$ iff $\{x\} \mathfrak{L}(S) \{y\}$ iff $x S y$. Hence $R = S$. ◀

► **Lemma 15.** $\mathfrak{L} : \mathbf{Rel} \rightarrow \mathbf{CABARel}$ is full. In other words, any directionally atomic $R : \mathcal{P}(X) \rightarrowtail \mathcal{P}(Y)$ is $\mathfrak{L}(S)$ for some relation $S : X \rightarrowtail Y$.

Proof. We will prove that $R = \mathfrak{L}(S)$ where $S : X \rightarrowtail Y$ is given by $x S y \stackrel{\text{def}}{=} \{x\} R \{y\}$.

Suppose that $A R B$. Then for any $a \in A$ we must have that $\{a\} R B$, as R is a bimodule. But then we must also have that there exists some $b \in B$ such that $\{a\} R \{b\}$, because R is atomic-founded, i.e. $a S b$. Therefore $A \mathfrak{L}(S) B$.

Suppose now that $A \mathfrak{L}(S) B$, which is to say that for all $a \in A$ there exists a $b \in B$ such that $\{a\} R \{b\}$. Because R is a bimodule this means that for all $a \in A$, $\{a\} R B$. But $A = \bigcup_{a \in A} \{a\}$, and therefore $A R B$, as R is left-disjunctive. ◀

To show that $\mathcal{L}$ is essentially surjective we define a functor $\mathbf{At} : \mathbf{CABARel} \rightarrow \mathbf{Rel}$ which inverts it:

$$\mathbf{At}(\mathcal{L}) \stackrel{\text{def}}{=} \{x \in \mathcal{L} \mid x \text{ is an atom of } \mathcal{L}\} \quad x \mathbf{At}(R) y \stackrel{\text{def}}{=} x R y$$

This operation restricts every relation to its atoms; it is evidently a functor.

► **Lemma 16.** $\mathcal{L} : \mathbf{Rel} \rightarrow \mathbf{CABARel}$ is essentially surjective.

Proof. Given a CABA $\mathcal{L}$ we define $R_{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{P}(\mathbf{At}(\mathcal{L}))$ in $\mathbf{CABARel}$ where $x R_{\mathcal{L}} X$ just if $x \sqsubseteq \bigsqcup X$. Conversely, we define $R_{\mathcal{L}}^{-1} : \mathcal{P}(\mathbf{At}(\mathcal{L})) \rightarrow \mathcal{L}$ by setting $X R_{\mathcal{L}}^{-1} x$ just if $\bigsqcup X \sqsubseteq x$. These relations are evidently natural, directionally atomic and inverses of each other. ◀

► **Theorem 17.** $\mathbf{Rel} \simeq \mathbf{CABARel}$.

Of course, Theorem 17 is not explicitly a duality; to make it one we have to compose it with the formal duality $(-)^{\dagger} : \mathbf{Rel}^{\text{op}} \simeq \mathbf{Rel}$ in the style of Kishida [21, §2.3], obtaining

► **Theorem 18** (Relational Tarski duality). $\mathbf{Rel}^{\text{op}} \simeq \mathbf{CABARel}$.

Intuitively, the fact that $\mathbf{Rel}$ is a dagger category means that its morphisms have no inherent ‘direction.’ Directionality appears when we restrict it to functions, making this equivalence a duality [21, §2.3] [25, Remark 4.1]. Indeed, it is possible to show that this duality is an ‘extension’ of the usual Tarski duality, in the sense that there is a commutative diagram

$$\begin{array}{ccc} \mathbf{Set}^{\text{op}} & \xleftarrow{\text{Gr}^{\text{op}}} & \mathbf{Rel}^{\text{op}} \\ \mathcal{P} \downarrow \simeq & & \simeq \downarrow \mathcal{L} \circ (-)^{\dagger} \\ \mathbf{CABA} & \xleftarrow{j} & \mathbf{CABARel} \end{array} \quad (1)$$

where Gr and j are faithful and injective-on-objects.

We define j by taking each CABA $\mathcal{B}$ to itself, and each complete Boolean homomorphism $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ to $j(f^*) : \mathcal{B} \rightarrow \mathcal{B}'$ given by $b j(f^*) b' \stackrel{\text{def}}{=} b \sqsubseteq f_!(b')$, where $f_! \dashv f^*$.

► **Lemma 19.** $j : \mathbf{CABA} \rightarrow \mathbf{CABARel}$ is faithful and injective-on-objects.

Proof. To begin we show that $j(f^*)$ is a morphism of $\mathbf{CABARel}$. First, $j(f^*)$ is left-disjunctive: suppose that for each $i \in I$ we have that $x_i j(f^*) y$, i.e. $x_i \sqsubseteq f_!(y)$. Then $\bigsqcup_{i \in I} x_i \sqsubseteq f_!(y)$, and hence $\bigsqcup_{i \in I} x_i j(f^*) y$. Second, $j(f^*)$ is atomic-founded. Suppose $x j(f^*) y$, i.e. $x \sqsubseteq f_!(y)$, with x an atom. Write $y = \bigsqcup_{i \in I} a_i$ for a_i atoms. Then $f_!(y) = \bigsqcup_{i \in I} f_!(a_i)$, because $f_!$ is a left adjoint and preserves joins. As x is an atom, $x \sqsubseteq f_!(a_i)$ for a particular $i \in I$, so $x j(f^*) a_i$ for some $a_i \sqsubseteq y$. Thirdly, $j(f^*)$ is a bimodule, by monotonicity of $f_!$.

j preserves identities: we have $x j(\text{id}) y$ iff $x \sqsubseteq \text{id}_!(y) = y$, viz. the identity in $\mathbf{CABARel}$. j preserves composition: let $f^* : X \rightarrow Y$ and $g^* : Y \rightarrow Z$ be complete Boolean homomorphisms. Then $x j(g^* \circ f^*) z$ iff $x \sqsubseteq f_!(g_!(z))$ iff $x j(f^*) g_!(z)$. But by reflexivity $g_!(z) \sqsubseteq g_!(z)$, i.e. $g_!(z) j(g^*) z$, so $x(j(f^*) ; j(g^*)) z$. For the converse, suppose $x j(f^*) y j(g^*) z$, i.e. $x \sqsubseteq f_!(y)$ and $y \sqsubseteq g_!(z)$. By the monotonicity of $f_!$ we have $x \sqsubseteq f_!(y) \sqsubseteq f_!(g_!(z))$, therefore $z j(g^* \circ f^*) x$.

Finally, j is evidently faithful by a Yoneda-type argument. ◀

We define Gr by taking each set X to itself, and each function $f : X \rightarrow Y$ to its graph $\text{Gr}(f) : X \rightarrow Y$, viz. $x \text{Gr}(f) y \stackrel{\text{def}}{=} (f(x) = y)$.

► **Lemma 20.** $\text{Gr} : \mathbf{Set} \rightarrow \mathbf{Rel}$ is faithful and injective-on-objects.

Proof. Gr clearly preserves identity and composition. It is faithful because functions are entirely determined by their graph. ◀

Finally, it is simple to calculate that (1) commutes.

280 **4.1 Sketch of a formal system**

281 Having developed this duality we can now see how it can be used for relating formulae.
 282 Suppose we have a relation $R : X \leftrightarrow Y$. Under the relational Tarski duality this induces a
 283 relation $\mathfrak{L}(R) : \mathcal{P}(X) \leftrightarrow \mathcal{P}(Y)$ between predicates. We denote this relation by the judgement

284
$$\varphi \xrightarrow{R} \psi$$

where $\varphi \in \mathcal{P}(X)$ is a predicate over X and $\psi \in \mathcal{P}(Y)$ over Y . Intuitively, this judgement says that *every state of X that satisfies φ is R -related to some state of Y that satisfies ψ* . The fact $\mathfrak{L}(R)$ is a bimodule means that the rule

$$\frac{\varphi' \vdash \varphi \quad \varphi \xrightarrow{R} \psi \quad \psi \vdash \psi'}{\varphi' \xrightarrow{R} \psi'}$$

285 is sound. This is reminiscent of the *consequence rule* of Hoare logic [14].

The fact $\mathfrak{L}(R)$ is left-disjunctive means that the rule

$$\frac{\varphi_1 \xrightarrow{R} \psi \quad \varphi_2 \xrightarrow{R} \psi}{(\varphi_1 \vee \varphi_2) \xrightarrow{R} \psi}$$

286 is sound. This allows us reasoning by cases on the left. The final characteristic property of
 287 $\mathfrak{L}(R)$, namely its atomic-foundedness, would require introducing judgements that capture
 288 atomicity of predicates, in the style of Abramsky [1].

Finally, the fact $\mathfrak{L}$ is a functor can be formally expressed by the rules

$$\frac{\varphi \xrightarrow{R} \psi \quad \psi \xrightarrow{S} \chi}{\varphi \xrightarrow{R;S} \chi} \quad \frac{\phi \vdash \psi}{\varphi \xrightarrow{\text{id}} \psi} \quad \frac{\varphi \xrightarrow{\text{id}} \psi}{\phi \vdash \psi}$$

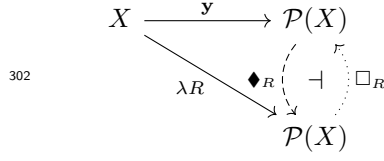
 289 **5 A Modal Relational Duality**

290 In this section we will extend the relational Tarski duality to (infinitary) classical modal
 291 logic. The basis for this is the *Thomason duality* [38] [21, §2.4], which uniquely associates
 292 each Kripke frame with a CABA equipped with a modal operator.

293 A *Kripke frame* (X, R) (also called a *transition system*) is a set X equipped with a
 294 relation $R : X \leftrightarrow X$. The intuition is that X is a set of *worlds*, and the relation R encodes
 295 *transitions* from one world to another. We will sometimes write $x \rightarrow_R y$ instead of $x_1 R x_2$.
 296 A morphism of Kripke frames $f : (X, R) \rightarrow (Y, S)$ is a function $f : X \rightarrow Y$ which preserves
 297 transitions, meaning $x_1 \rightarrow_R x_2$ implies $f(x_1) \rightarrow_S f(x_2)$. Kripke frames and their morphisms
 298 form a category **Frm**.

299 The relation R can also be seen as a map $\lambda R : X \rightarrow \mathcal{P}(X)$ by currying. Then there is
 300 a unique join-preserving function $\blacklozenge_R : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ that makes the following diagram

commute, where $\mathbf{y} : X \rightarrow \mathcal{P}(X)$ maps every element $x \in X$ to the singleton set $\{x\}$.⁵



As $\diamond_R$ preserves joins and $\mathcal{P}(X)$ is a complete lattice, the adjoint functor theorem implies that it has a right adjoint $\square_R : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$, which preserves all meets.

These maps are explicitly given by

$$\diamond_R(A) = \{w \in X \mid \exists v \in A. v R w\} \quad \square_R(A) = \{w \in X \mid \forall v \in X. w R v \implies v \in A\}$$

Having one of these three pieces of data (the relation R ; a join-preserving map $\diamond_R$; or a meet-preserving map $\square_R$) uniquely determines the other two [20].

It is possible to define morphisms of frames purely in terms of operators:

► **Lemma 21.** *A function $f : X \rightarrow Y$ is a morphism of frames $f : (X, R) \rightarrow (Y, S)$ if and only if $f^* \circ \square_S \subseteq \square_R \circ f^*$.*

Let $\mathbf{CABAO}^-$ be the category whose objects $(\mathcal{B}, \square_{\mathcal{B}})$ are CABAs $\mathcal{B}$ equipped with an operator $\square : \mathcal{B} \rightarrow \mathcal{B}$ that preserves all meets,⁶ and morphisms $f^* : (\mathcal{B}, \square_{\mathcal{B}}) \rightarrow (\mathcal{B}', \square_{\mathcal{B}'})$ are complete Boolean homomorphisms $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ that satisfy $f^* \circ \square_{\mathcal{B}} \subseteq \square_{\mathcal{B}'} \circ f^*$ (with the pointwise order). This yields a modal duality

► **Theorem 22** (Weak Thomason duality). $\mathbf{Frm}^{\text{op}} \simeq \mathbf{CABAO}^-$.

The maps of CABAOs preserve the Boolean structure, but they only preserve $\square$ weakly. To strengthen this we need the notion of an *open map*.

► **Definition 23.** *A morphism of frames $f : (X, R) \rightarrow (Y, S)$ is open just if $f(x) \rightarrow_S y'$ implies that there exists an $x' \in X$ with $x \rightarrow_R x'$ and $f(x') = y'$.*

We let $\mathbf{Frm}_{\text{open}}$ be the wide subcategory of $\mathbf{Frm}$ whose morphisms are open. It is easy to show that such an f is open iff $f^* \circ \square_S = \square_R \circ f^*$. Then, letting $\mathbf{CABAO}$ be the wide subcategory of $\mathbf{CABAO}^-$ whose morphisms f^* preserve $\square$ (i.e. $f^* \circ \square = \square \circ f^*$) we obtain a refinement of Theorem 22:

► **Theorem 24** (Thomason duality). $\mathbf{Frm}_{\text{open}}^{\text{op}} \simeq \mathbf{CABAO}$.

To obtain a relational version of this duality we must relax the functionality of morphisms of Kripke frames. To achieve that, notice that the notion of an open map is precisely a *functional bisimulation* [34, §3.2]. Hence, we replace open maps with general *(bi)simulations*.

► **Definition 25** (Simulations, Cosimulations, and Bisimulations). *Let (X, R) and (Y, S) be Kripke frames, and let $Q : X \rightarrowtail Y$ be a relation.*

1. Q is a simulation when for any $x Q y$ and $x \rightarrow_R x'$ we have a $y' \in Y$ with $y \rightarrow_S y'$ and $x' Q y'$.
2. Q is a cosimulation whenever $Q^\dagger$ is a simulation.

⁵ This is also known as the *left Kan extension* of λR along the Yoneda embedding.

⁶ Or, equivalently, a join-preserving operator $\diamond : \mathcal{B} \rightarrow \mathcal{B}$.

334 3. Q is a bisimulation if it is both a simulation and a cosimulation.

335 We illustrate the definitions of simulation and cosimulation pictorially:

$$\begin{array}{ccc}
 x' & \dashrightarrow^Q & \exists y' \\
 R \uparrow & & \uparrow S \\
 x & \xrightarrow{Q} & y
 \end{array}
 \qquad
 \begin{array}{ccc}
 \exists x' & \dashrightarrow^Q & y' \\
 R \uparrow & & \uparrow S \\
 x & \xrightarrow{Q} & y
 \end{array}$$

337 These two conditions are often called the *forth* and the *back* condition.

338 It is easy to see that an open map is exactly a function whose graph is a bisimulation. Indeed, the fact it preserves transitions means it is a simulation (i.e. it satisfies the ‘forth’ condition). The fact it is open means it is a cosimulation (i.e. it satisfies the ‘back’ condition).

339 (Co)simulations are closed under relational composition, and that the identity relation is a bisimulation. We thus obtain two categories **FrmSim** and **FrmBisim** with Kripke frames as objects, and simulations and bisimulations as morphisms respectively. Noting that the opposite of a simulation is a cosimulation, and vice versa. Therefore, there is a formal duality $(-)^{\dagger} : \mathbf{FrmBisim}^{\text{op}} \simeq \mathbf{FrmBisim}$, meaning **FrmBisim** is also a dagger category.

340 It is possible to characterise simulations purely in terms of the lower relation.

341 ► **Lemma 26.** *A relation $Q : X \rightarrowtail Y$ is a simulation $Q : (X, R) \rightarrowtail (Y, S)$ if and only if $A \mathfrak{L}(Q) B$ implies $\blacklozenge A \mathfrak{L}(Q) \blacklozenge B$ for any $A \subseteq X$ and $B \subseteq Y$.*

342 **Proof.** Suppose that $Q : (X, R) \rightarrowtail (Y, S)$ is a simulation and that $A \mathfrak{L}(Q) B$. If $x \in \blacklozenge A$, then there exists some $a \in A$ with $a \rightarrow_R x$. We know that there exists some $b \in B$ for which $a Q b$, so by simulation there must exist some $y \in Y$ such that $b \rightarrow_S y$, so $y \in \blacklozenge B$, and $x Q y$.

343 For the converse, suppose that if $A \mathfrak{L}(Q) B$ then $\blacklozenge A \mathfrak{L}(Q) \blacklozenge B$, and that $x Q y$. Then we know that $\{x\} \mathfrak{L}(Q) \{y\}$ and therefore $\blacklozenge \{x\} \mathfrak{L}(Q) \blacklozenge \{y\}$. So, for any element $x' \in X$ with $x \rightarrow_R x'$ we know that $x' \in \blacklozenge \{x\}$, therefore there exists some $y' \in \blacklozenge \{y\}$ —that is, $y' \in Y$ with $y \rightarrow_S y'$ —for which $x' Q y'$. ◀

344 It is also possible to state this property in what we call a ‘half-adjoint’ style.

345 ► **Lemma 27.** *Let (X, R) and (Y, S) be Kripke frames, and $Q : X \rightarrowtail Y$. Then Q is a simulation $(X, R) \rightarrowtail (Y, S)$ iff $A \mathfrak{L}(Q) \Box B$ implies $\blacklozenge A \mathfrak{L}(Q) B$.*

346 **Proof.** For the forwards direction suppose that $A \mathfrak{L}(Q) \Box B$. Because Q is a simulation we have $\blacklozenge A \mathfrak{L}(Q) \blacklozenge \Box B$ by Lemma 26. But $\blacklozenge \neg \Box$ so $\blacklozenge \Box B \subseteq B$, and hence $(\blacklozenge A) \mathfrak{L}(Q) B$ as $\mathfrak{L}(Q)$ is a bimodule.

347 For the backwards direction suppose $A \mathfrak{L}(Q) B$. Notice that as $\blacklozenge \neg \Box$ we have $B \subseteq \Box \blacklozenge B$. As $\mathfrak{L}(Q)$ is a bimodule we know that $A \mathfrak{L}(Q) (\Box \blacklozenge B)$. Then by assumption $\blacklozenge A \mathfrak{L}(Q) \blacklozenge B$, which by Lemma 26 implies that Q is a simulation. ◀

348 ► **Definition 28.** *A simulatory relation $Q : (X, \Box_R) \rightarrowtail (Y, \Box_S)$ between CBAOs is a directionally atomic $Q : X \rightarrowtail Y$ for which the condition of Lemma 27 holds, i.e.*

$$349 \quad A Q \Box_S B \implies \blacklozenge_R A Q B.$$

350 ► **Proposition 29.** *If $R : (X, \Box_X) \rightarrowtail (Y, \Box_Y)$ and $S : (Y, \Box_Y) \rightarrowtail (Z, \Box_Z)$ are simulatory relations, then their relational composition is a simulatory relation $R;S : (X, \Box_X) \rightarrowtail (Z, \Box_Z)$.*

351 Define **CABAOSim** to have CABAs with operators $(\mathcal{B}, \Box_{\mathcal{B}})$ as objects and simulatory relations $Q : (X, \Box_X) \rightarrowtail (Y, \Box_Y)$ as morphisms. We will often write these morphisms simply as $Q : X \rightarrowtail Y$ and omit subscripts. Using the preceding lemma it is easy to check that this is a category, with $\Box$ as an identity morphism.

374 ► **Lemma 30.** $\mathcal{L}$ is a functor $\mathbf{FrmSim} \rightarrow \mathbf{CABAOSim}$.

375 ► **Lemma 31.** $\mathcal{L} : \mathbf{FrmSim} \rightarrow \mathbf{CABAOSim}$ is faithful.

376 ► **Lemma 32.** $\mathcal{L} : \mathbf{FrmSim} \rightarrow \mathbf{CABAOSim}$ is full. In other words, every simulatory
377 relation $Q : (\mathcal{P}(X), \sqsubseteq_R) \leftrightarrow (\mathcal{P}(Y), \sqsubseteq_S)$ is $\mathcal{L}(T)$ for some simulation $T : (X, R) \leftrightarrow (Y, S)$.

378 We can define a functor $\mathbf{At} : \mathbf{CABAOSim} \rightarrow \mathbf{FrmSim}$ by restricting a simulatory
379 relation to atoms, as before. By Lemma 32 it is clearly a functor.

380 ► **Lemma 33.** $\mathcal{L} : \mathbf{FrmSim} \rightarrow \mathbf{CABAOSim}$ is essentially surjective.

381 **Proof.** Following the proof of Lemma 16, given a CABAO $\mathcal{B}$, define the relations $R_{\mathcal{B}} : \mathcal{B} \leftrightarrow$
382 $\mathcal{P}(\mathbf{At}(\mathcal{B}))$ and $R_{\mathcal{B}}^{-1} : \mathcal{P}(\mathbf{At}(\mathcal{B})) \leftrightarrow \mathcal{B}$ by $x R_{\mathcal{B}} X$ just if $x \sqsubseteq \bigsqcup X$ and $X R_{\mathcal{B}}^{-1} c$ just if $\bigsqcup X \sqsubseteq c$
383 respectively. As in the proof of Lemma 16 these are both directionally atomic and evidently
384 inverses to each other. Thus, it suffices to show that they are simulatory.

385 Suppose that $x \sqsubseteq \bigsqcup \square X$, then

$$386 \quad x \sqsubseteq \bigsqcup \square X \sqsubseteq \square \blacklozenge \bigsqcup \square X = \square \bigsqcup \blacklozenge \square X \sqsubseteq \square \bigsqcup X$$

387 by properties of the adjunction $\blacklozenge \dashv \square$. Hence $\blacklozenge x \sqsubseteq \bigsqcup X$, and $R_{\mathcal{B}}$ is simulatory.

388 Similarly, if $\bigsqcup X \sqsubseteq \square x$, then $\blacklozenge \bigsqcup X \sqsubseteq x$ and therefore $\bigsqcup \blacklozenge X \sqsubseteq x$, so $R_{\mathcal{B}}^{-1}$ is simulatory.

389 ◀

390 In summary, we obtain an equivalence

391 ► **Theorem 34.** $\mathbf{FrmSim} \simeq \mathbf{CABAOSim}$.

392 As in our previous relational duality, the directionality disappears in the relational case.
393 We could turn this result into a duality by composing it with a formal duality between
394 $\mathbf{FrmSim}$ and the category of cosimulations. However, this is somewhat awkward.

395 Our final objective is to restrict this equivalence by analogy to the restriction of
396 $\mathbf{Frm}^{\text{op}} \simeq \mathbf{CABAO}^-$ to $\mathbf{Frm}_{\text{open}}^{\text{op}} \simeq \mathbf{CABAO}$, where the maps in $\mathbf{Frm}_{\text{open}}^{\text{op}}$ are open, and
397 correspondingly the maps in $\mathbf{CABAO}$ preserve all logical connectives. In our relational
398 duality the morphisms are no longer functional, but merely simulations. Thus, openness will
399 turn them into *bisimulations*.

400 Unfortunately, expressing bisimilarity using our current vocabulary is not immediately
401 possible. For that we will need the following ‘dual’ relational lifting.

402 ► **Definition 35 (Upper Relation).** Given a relation $R : X \leftrightarrow Y$, the associated upper relation
403 $\mathfrak{U}(R) : \mathcal{P}(X) \leftrightarrow \mathcal{P}(Y)$ is defined by

$$404 \quad S \mathfrak{U}(R) T \stackrel{\text{def}}{=} \forall t \in T. \exists s \in S. s R t.$$

405 It is easy to see that $\mathfrak{U}(R) = \mathcal{L}(R^\dagger)^\dagger$ for any $R : X \leftrightarrow Y$. This makes the theory dual, in the
406 sense that $\mathfrak{U}(R)$ is *opdirectionally atomic* (i.e. right-disjunctive, atomic op-founded, and an
407 opbimodule). To avoid this proliferation of concepts we will simply work with the opposite
408 relation $\mathcal{L}(R^\dagger)$. We can construct this relation directly on relations between CABAs.

409 ► **Definition 36.** For a relation $R : \mathcal{B} \leftrightarrow \mathcal{B}'$ between CABAs we define its variant relation
410 $R^\bullet : \mathcal{B}' \leftrightarrow \mathcal{B}$ to be

$$411 \quad b' R^\bullet b \stackrel{\text{def}}{=} \forall \text{atom } a' \sqsubseteq b'. \exists \text{atom } a \sqsubseteq b. a R a'.$$

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This acts in the expected way between lifted relations.

► **Lemma 37.** $\mathfrak{L}(R)^\bullet = \mathfrak{L}(R^\dagger)$ for any $R : X \leftrightarrow Y$, and hence $\mathfrak{U}(R) = \mathfrak{L}(R)^\bullet^\dagger$.

Proof. Let $R : X \leftrightarrow Y$ be a relation between sets. Then for any $T \subseteq Y$ and $S \subseteq X$ we have $T \mathfrak{L}(R)^\bullet S$ just if $\forall s \in S. \exists t \in T. t R s$. Note that this is exactly the same as $T \mathfrak{L}(R^\dagger) S$. ◀

As every directionally atomic relation is in the image of $\mathfrak{L}$, Lemma 37 implies that the variant of a directionally atomic relation is directionally atomic. Hence, the variant construction forms a functor $(-)^\bullet : \mathbf{CABARel}^{\text{op}} \rightarrow \mathbf{CABARel}$ which sends every CABA to itself, and every directionally atomic relation to its variant. Moreover,

► **Theorem 38.** $(-)^\bullet : \mathbf{CABARel}^{\text{op}} \rightarrow \mathbf{CABARel}$ is an equivalence.

Thus, we have a self-duality on $\mathbf{CABARel}$, which makes the following claims straightforward.

Like with simulations and Lemma 26, it is possible to characterise cosimulations purely in terms of the variant relation.

► **Lemma 39.** A relation $Q : X \leftrightarrow Y$ is a cosimulation $Q : (X, R) \leftrightarrow (Y, S)$ if and only if $B \mathfrak{L}(Q)^\bullet A$ implies $\blacklozenge B \mathfrak{L}(Q)^\bullet \blacklozenge A$ for any $B \subseteq Y$ and $A \subseteq X$.

Proof. R is a cosimulation iff $R^\dagger$ is a simulation, so use Lemmas 26 and 37. ◀

We can also characterise this condition as a ‘half-adjunction’ (cf. Lemma 27).

► **Lemma 40.** Let (X, R) and (Y, S) be Kripke frames, and $Q : X \leftrightarrow Y$. Then Q is a cosimulation $(X, R) \leftrightarrow (Y, S)$ iff $B \mathfrak{L}(Q)^\bullet \square A$ implies $\blacklozenge B \mathfrak{L}(Q)^\bullet A$.

The proof is completely dual to Lemma 27.

Define a *cosimulatory relation* $Q : (X, \square_R) \leftrightarrow (Y, \square_S)$ between CABAOs to be a directionally atomic relation $Q : X \leftrightarrow Y$ for which the condition of Lemma 40 holds, i.e. $B Q^\bullet \square_R A$ implies $\blacklozenge_S B Q^\bullet A$. The following proposition is analogous to Proposition 29.

► **Proposition 41.** If $R : (X, \square_X) \leftrightarrow (Y, \square_Y)$ and $S : (Y, \square_Y) \leftrightarrow (Z, \square_Z)$ are cosimulatory relations, then their composition is a cosimulatory relation $R ; S : (X, \square_X) \leftrightarrow (Z, \square_Z)$.

Call a relation *bisimulatory* if it is both simulatory and cosimulatory. Define $\mathbf{CABAOBisim}$ to have CABAOs $(\mathcal{B}, \square_{\mathcal{B}})$ as objects and bisimulatory relations $Q : (X, \square_R) \leftrightarrow (Y, \square_S)$ as morphisms. This is again a category with $\sqsubseteq$ as an identity morphism.

► **Lemma 42.** $\mathfrak{L}$ is a functor $\mathbf{FrmBisim} \rightarrow \mathbf{CABAOBisim}$.

► **Lemma 43.** $\mathfrak{L} : \mathbf{FrmBisim} \rightarrow \mathbf{CABAOBisim}$ is faithful.

► **Lemma 44.** $\mathfrak{L} : \mathbf{FrmBisim} \rightarrow \mathbf{CABAOBisim}$ is full. In other words, every simulatory relation $Q : (\mathcal{P}(X), \square_R) \leftrightarrow (\mathcal{P}(Y), \square_S)$ is $\mathfrak{L}(T)$ for some bisimulation $T : (X, R) \leftrightarrow (Y, S)$.

► **Lemma 45.** $\mathfrak{L} : \mathbf{FrmBisim} \rightarrow \mathbf{CABAOBisim}$ is essentially surjective.

Proof. We augment the proof of Lemma 33 by showing that both $R_{\mathcal{B}} : \mathcal{B} \leftrightarrow \mathcal{P}(\text{At}(\mathcal{B}))$ and $R_{\mathcal{B}}^{-1} : \mathcal{P}(\text{At}(\mathcal{B})) \leftrightarrow \mathcal{B}$ given by $x R_{\mathcal{B}} X$ just if $x \sqsubseteq \bigsqcup X$ and $X R_{\mathcal{B}}^{-1} c$ just if $\bigsqcup X \sqsubseteq x$ respectively are also cosimulatory.

First, see that $X R_{\mathcal{B}}^\bullet x$ just if for any atom $A \sqsubseteq X$ there exists an atom $a \sqsubseteq x$ for which $a \sqsubseteq \bigsqcup A$. But by atomicity, this is the same as $a = A$, so $X R_{\mathcal{B}}^\bullet x$ just if $X \sqsubseteq x$. Hence it is obvious that $X \sqsubseteq \square x$ implies $\blacklozenge X \sqsubseteq x$, and $R_{\mathcal{B}}$ is cosimulatory.

Second, see that $x R_{\mathcal{B}}^{-1 \bullet} X$ just if for any atom $a \sqsubseteq x$ there exists an atom $A \sqsubseteq X$ for which $\bigsqcup A \sqsubseteq a$. But by atomicity this is just $A = \{a\}$, so $x R_{\mathcal{B}}^{-1 \bullet} X$ just if $x \sqsubseteq X$. Hence it is obvious that $x \sqsubseteq \square X$ implies $\blacklozenge x \sqsubseteq X$, and $R_{\mathcal{B}}^{-1}$ is cosimulatory. ◀

453 We thus obtain the following equivalence.

454 ► **Theorem 46.** $\mathbf{FrmBisim} \simeq \mathbf{CABAOBisim}$.

455 As before, this is not explicitly a duality, but we can obtain one from the formal duality
456 $(-)^{\dagger} : \mathbf{FrmBisim}^{\text{op}} \simeq \mathbf{FrmBisim}$.

457 ► **Theorem 47** (Relational Thomason duality). $\mathbf{FrmBisim}^{\text{op}} \simeq \mathbf{CABAOBisim}$.

458 We can show that Theorem 47 is an extension of the Thomason duality; as with (1),
459 there is a commutative diagram of functors

$$\begin{array}{ccc}
 \mathbf{Frm}_{\text{open}}^{\text{op}} & \xleftarrow{\text{Gr}^{\text{op}}} & \mathbf{FrmBisim}^{\text{op}} \\
 \mathcal{P} \downarrow \simeq & & \simeq \downarrow \mathcal{L} \circ (-)^{\dagger} \\
 \mathbf{CABAO} & \xrightarrow{j} & \mathbf{CABAOBisim}
 \end{array} \tag{2}$$

461 where Gr and j are faithful and injective-on-objects.

462 We define j as before, by taking each CABAO $(\mathcal{B}, \square_{\mathcal{B}})$ to itself, and each complete
463 Boolean homomorphism $f^* : \mathcal{B} \rightarrow \mathcal{B}'$ to $j(f^*) : \mathcal{B} \rightarrow \mathcal{B}'$ given by $b j(f^*) b' \stackrel{\text{def}}{=} b \sqsubseteq f_!(b')$.

464 ► **Lemma 48.** $j : \mathbf{CABAO} \rightarrow \mathbf{CABAOBisim}$ is a faithful functor.

465 **Proof.** By extension of Lemma 19, we know that $j(f^*)$ is a directionally atomic relation,
466 and that j preserves identities and composition, and that it is faithful. It remains only to
467 show that $j(f^*)$ is bisimulatory.

468 First, to show that $j(f^*)$ is simulatory, suppose that $x j(f^*) (\square y)$, that is, $x \sqsubseteq f_!(\square y)$.
469 Then

$$470 \quad x \sqsubseteq f_!(\square y) \sqsubseteq \square \blacklozenge f_!(\square y) \sqsubseteq \square f_!(\blacklozenge \square y) \sqsubseteq \square f_!(y)$$

471 by the properties of the adjunction $\blacklozenge \dashv \square$. Hence $\blacklozenge x \sqsubseteq f_!(y)$, so $j(f^*)$ is simulatory.

472 Second, to show that $j(f^*)$ is cosimulatory, notice first that $b j(f^*) a$ just if, for any atom
473 $y \sqsubseteq b$ there exists some atom $x \sqsubseteq a$ for which $x \sqsubseteq f_!(y)$, which is the same as $x = f_!(y)$ by
474 the properties of atoms and recalling that $f_!$ preserves atoms. Therefore, $b j(f^*) a$ just if
475 $b \sqsubseteq f^*(a)$. Hence it is easy to see that, if $b \sqsubseteq f^*(\square a) = \square f^*(a)$, then $\blacklozenge b \sqsubseteq f^*(a)$, and $j(f^*)$
476 is cosimulatory. ◀

477 As usual, Gr takes each frame (X, R) to itself and each open map $f : (X, R) \rightarrow (Y, S)$ to
478 the graph $\text{Gr}(f) : X \rightarrow Y$ defined by $x \text{Gr}(f) y \stackrel{\text{def}}{=} f(x) = y$.

479 ► **Lemma 49.** The functor $\text{Gr} : \mathbf{Frm}_{\text{open}} \rightarrow \mathbf{FrmBisim}$ is faithful.

480 **Proof.** Gr clearly takes open maps to bisimulations, preserves identity and composition. It
481 is faithful because any function is determined entirely by its graph. ◀

482 Finally, it is simple to calculate that (2) commutes.

483 5.1 Sketch of a formal system

Recall the formal system of the judgment $\searrow^Q$ for a relation $Q : X \rightarrow Y$ that we sketched
in Section 4. Now assume that (X, R) and (Y, S) are Kripke frames. We can extend this

system to a modal logic where the logic over X and Y have corresponding modalities $\blacklozenge$ and $\Box$. If Q is a simulation then by Lemmas 26 and 27 we can extend the system with the rules

$$\frac{\varphi \xrightarrow{R} \psi}{\blacklozenge \varphi \xrightarrow{R} \blacklozenge \psi} \qquad \frac{\varphi \xrightarrow{R} \Box \psi}{\blacklozenge \varphi \xrightarrow{R} \psi}$$

If R is a cosimulation we can use Lemmas 39 and 40 to add the rules

$$\frac{\varphi \xrightarrow{R^\dagger} \psi}{\blacklozenge \varphi \xrightarrow{R^\dagger} \blacklozenge \psi} \qquad \frac{\varphi \xrightarrow{R^\dagger} \Box \psi}{\blacklozenge \varphi \xrightarrow{R^\dagger} \psi}$$

5.2 A worked example: bisimulation between buffer systems

We will now show how to use these rules to reason over a particular bisimulation. The two frames both model a *buffer* holding a single natural number. However, the two buffers have *internal branching*: the first has two different ‘ways’ of holding a number (the left and right slots), while the second one has three (slots A, B, and C). This might happen if the buffer is provided by a distributed cluster of machines whose internal structure is not observable.

Define $(X, \rightarrow_X)$ by letting $X = \{\text{empty}_X\} \cup \{L(n) \mid n \in \mathbb{N}\} \cup \{R(n) \mid n \in \mathbb{N}\}$ with

$$\text{empty}_X \rightarrow_X L(n), \quad \text{empty}_X \rightarrow_X R(n), \quad L(n) \rightarrow_X \text{empty}_X, \quad R(n) \rightarrow_X \text{empty}_X$$

for all $n \in \mathbb{N}$. Similarly, define $(Y, \rightarrow_Y)$ by $Y = \{\text{empty}_Y\} \cup \{A(n), B(n), C(n) \mid n \in \mathbb{N}\}$ with

$$\text{empty}_Y \rightarrow_Y A(n) \quad \text{empty}_Y \rightarrow_Y B(n) \quad \text{empty}_Y \rightarrow_Y C(n)$$

$$A(n) \rightarrow_Y \text{empty}_Y \quad B(n) \rightarrow_Y \text{empty}_Y \quad C(n) \rightarrow_Y \text{empty}_Y$$

for all $n \in \mathbb{N}$. There is a bisimulation $Q : X \rightarrowtail Y$ between these frames given by

$$Q = \{(\text{empty}_X, \text{empty}_Y)\} \cup \{(w, v) \mid n \in \mathbb{N}, w \in F_X(n), v \in F_Y(n)\}$$

where $F_X(n) = \{L(n), R(n)\}$ and $F_Y(n) = \{A(n), B(n), C(n)\}$. In other words, full states holding the same value are related, and so are empty states. This is evidently a non-functional bisimulation.

We define the following predicates, parametrised by $n \in \mathbb{N}$. On X let $\text{empty}_X = \{\text{empty}_X\}$, $L_n = \{L(n)\}$, and $R_n = \{R(n)\}$. On Y let $\text{empty}_Y = \{\text{empty}_Y\}$.

We will use the following facts as the ‘background theory’:

$$\blacklozenge \text{empty}_X = \bigvee_n F_X(n) \qquad \Box \text{empty}_X = \bigvee_n F_X(n) \qquad L_n \xrightarrow{Q} F_Y(n)$$

$$\blacklozenge \text{empty}_Y = \bigvee_n F_Y(n) \qquad \Box \text{empty}_Y = \bigvee_n F_Y(n) \qquad R_n \xrightarrow{Q} F_Y(n)$$

The first two columns arise by direct observation. The last column holds because every left/right state is simulated by a full state holding the same value.

We derive the fact that if the previous state in the first buffer was a left or right one holding n , then the current state is simulated by an empty state in the second buffer:

$$(\blacklozenge L_n \vee \blacklozenge R_n) \xrightarrow{Q} \text{empty}_Y$$

The derivation proceeds in two symmetric branches, combined by disjunction:

$$\begin{array}{c}
 \frac{L_n \xrightarrow{Q} F_Y(n) \quad F_Y(n) \vdash \Box \text{empty}_Y}{L_n \xrightarrow{Q} \Box \text{empty}_Y} \quad \frac{R_n \xrightarrow{Q} F_Y(n) \quad F_Y(n) \vdash \Box \text{empty}_Y}{R_n \xrightarrow{Q} \Box \text{empty}_Y} \\
 \hline
 \frac{\frac{L_n \xrightarrow{Q} \Box \text{empty}_Y} \quad \frac{R_n \xrightarrow{Q} \Box \text{empty}_Y}}{\Diamond L_n \xrightarrow{Q} \text{empty}_Y \quad \Diamond R_n \xrightarrow{Q} \text{empty}_Y} \\
 \hline
 (\Diamond L_n \vee \Diamond R_n) \xrightarrow{Q} \text{empty}_Y
 \end{array}$$

6 Related work

A few relational dualities have been previously described in the literature. Most are of the form $\mathcal{C}^{\text{op}} \simeq \mathcal{D}$ where $\mathcal{C}$ is a category whose morphisms are relations, whereas $\mathcal{D}$ is a category of algebras with some form of *hemimorphism*, i.e. a morphism preserving most—but not all!—the logical structure. The earliest duality of the form is $\mathbf{Rel} \simeq \mathbf{CABA}_{\vee}$. This duality has been rediscovered multiple times, but is likely due to Jónsson [19]. Kishida [21, §2.3] argues that this extends to a ‘2-duality,’ as both of these are (strict) 2-categories. Halmos [10] extended it to the continuous case, i.e. a duality between Stone spaces with continuous relations on the one hand, and Boolean algebras with hemimorphisms on the other. Cignoli et al. [5] do something similar for Priestly spaces and continuous monotone relations. Hofmann and Nora [15] have proposed a general framework for such dualities, obtaining the relational side as the Kleisli category of a suitable monad. In later work they extended such dualities to metric structures and quantale-enriched categories [16].

Kurz, Moshier, and Jung [25] present dualities that are much closer to the flavour we employ in this paper. They achieve this by working in an order-enriched setting, where relations can be presented as both spans and cospans in a 2-categorical manner. They use this to extend (well-behaved) dualities to dualities between categories of relations, and even adjunctions of categories to adjunctions between framed bicategories [36]. For example, if their recipe is applied to the category $\mathbf{Pos}$ of posets and monotone functions it lifts a bimodule $R : X \leftrightarrow Y$ to the relation $2^R : [X, 2] \leftrightarrow [Y, 2]$ between upper sets that is defined by letting $A \ 2^R \ B$ just if $x \in A$ and $x R y$ implies $y \in B$. Hence, the type of lifting they obtain is very different, and it is not apparent that it is related to bisimulations.

Birkmann, Urbat, and Milius [2] present extensions of categorical dualities through monoidal adjunctions and apply them to algebraic language theories. As a corollary they obtain some relational dualities, e.g. between well-behaved relations of profinite ordered monoids and natural morphisms of residuation algebras.

Malacaria [29] presents the Thomason duality in a new light, and shows how it can be used to give bisimulation an algebraic meaning. In particular, the main theorem shows that two Kripke frames are bisimilar just if their associated (dual) algebras have an isomorphic subalgebra.

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